Section 3: UML Diagrams — SpamGuard AI

Evaluation: Software Engineering Internal Assessment 3 (IA3)
Academic Micro-Project: Micro Project – 05 (Semester 3)
System Name: SpamGuard AI — AI-Based Spam Email Classifier
Document Classification: Section 3 Compliance Artifact & Software Architecture Design
Release Date: October 2026
Status: 100% Implemented & UML Standard Compliant

Overview of Architectural UML Diagrams

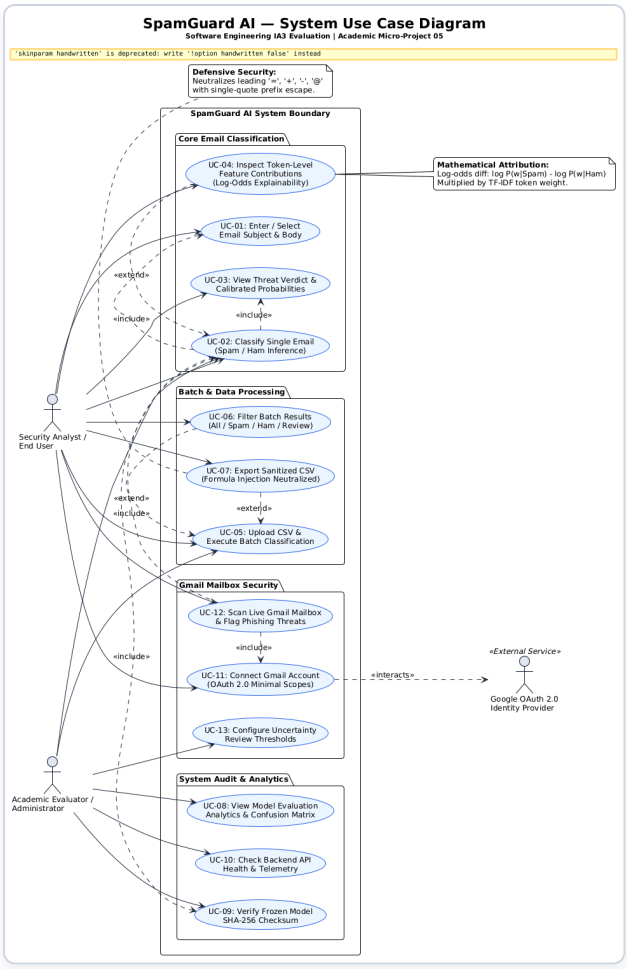
In strict accordance with Section 3 of the IA3 Guidelines, exactly five standard UML diagrams have been developed to model the static structure, dynamic behavior, user interactions, and object lifecycles of SpamGuard AI:

1. **Use Case Diagram (`uml/01_use_case.puml`):** Models system actors, system boundary, and user capabilities with `<<include>>` and `<<extend>>` associations.
2. **Class Diagram (`uml/02_class_diagram.puml`):** Models core domain classes (`SpamPredictor`, `PredictionResult`, `FeatureContribution`), FastAPI Pydantic request/response schemas, exception hierarchies, and Scikit-learn pipeline bindings.
3. **Sequence Diagram (`uml/03_sequence_diagram.puml`):** Models the end-to-end runtime message exchange for a single-email classification across Next.js UI, FastAPI, Preprocessor, Scikit-learn Pipeline, and Log-Odds Explainer.
4. **Activity Diagram (`uml/04_activity_diagram.puml`):** Represents the end-to-end classification workflow, input normalization, boundary validation decision branches, model inference, and explainability extraction.
5. **State Machine Diagram (`uml/05_state_machine.puml`):** Models the finite state lifecycle of an email scan from `Idle` through `InputReady`, `Validating`, `Processing`, `ResultAvailable`, and defensive error states.

All source `.puml` files and rendered high-resolution `.png` and `.svg` files are located in `docs/ia3/uml/`.

Diagram 1: System Use Case Diagram

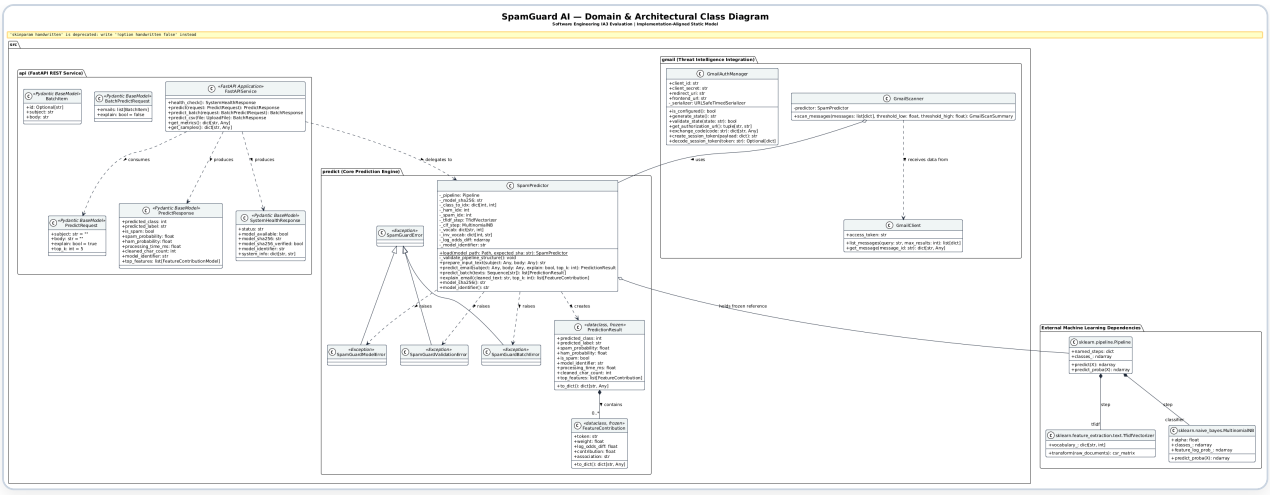
- **Source File:** `docs/ia3/uml/01_use_case.puml`
- **Rendered Artifacts:** `01_use_case.png` | `01_use_case.svg`
- **Description:** Illustrates the interactions between the primary human actors (**Security Analyst / End User**, **Academic Evaluator / Administrator**) and external systems (**Google OAuth 2.0 Identity Provider**) within the SpamGuard AI boundary. It captures core capabilities including single email classification, mathematical explainability, CSV batch classification, formula injection neutralization, model integrity audits, and live Gmail scanning.



SpamGuard AI Use Case Diagram

Diagram 2: Domain & Class Diagram

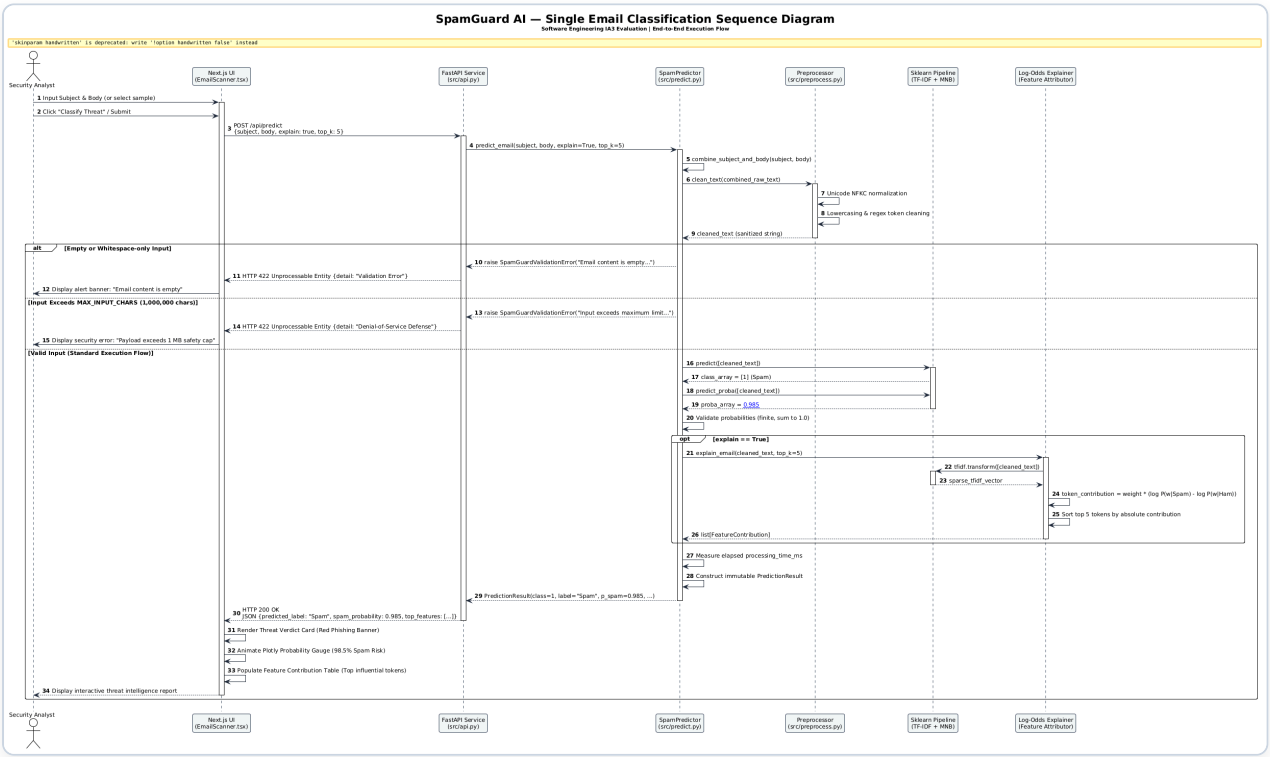
- **Source File:** `docs/ia3/uml/02_class_diagram.puml`
- **Rendered Artifacts:** `02_class_diagram.png` | `02_class_diagram.svg`
- **Description:** Details the static structure of the codebase, reflecting the exact classes implemented in `src/predict.py`, `src/api.py`, and `src/gmail/`. Shows attribute types, method signatures, dataclass compositions (`PredictionResult` aggregates `FeatureContribution`), custom exception inheritance (`SpamGuardError` hierarchy), and external machine learning dependencies (`Pipeline`, `TfidfVectorizer`, `MultinomialNB`).



SpamGuard AI Class Diagram

Diagram 3: Single Email Classification Sequence Diagram

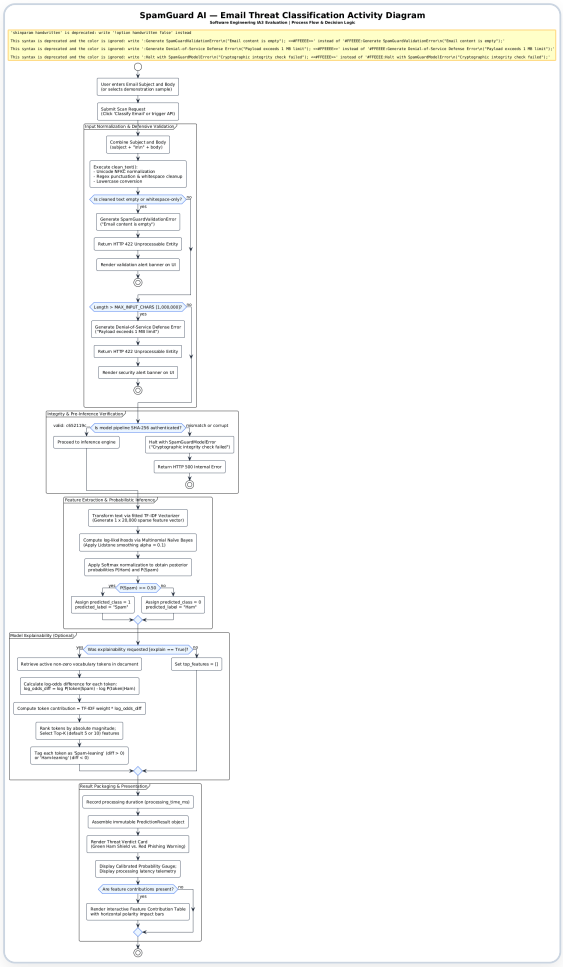
- **Source File:** `docs/ia3/uml/03_sequence_diagram.puml`
- **Rendered Artifacts:** `03_sequence_diagram.png` | `03_sequence_diagram.svg`
- **Description:** Traces the chronological order of message execution during a single email scan: User \to Next.js UI \to FastAPI REST endpoint \to `SpamPredictor` \to `clean_text` normalization \to Scikit-learn Pipeline \to Log-Odds Explainer \to UI Threat Rendering. It models both positive classification and alternative validation error paths (empty text and 1MB DoS violations).



SpamGuard AI Sequence Diagram

Diagram 4: Threat Classification Activity Diagram

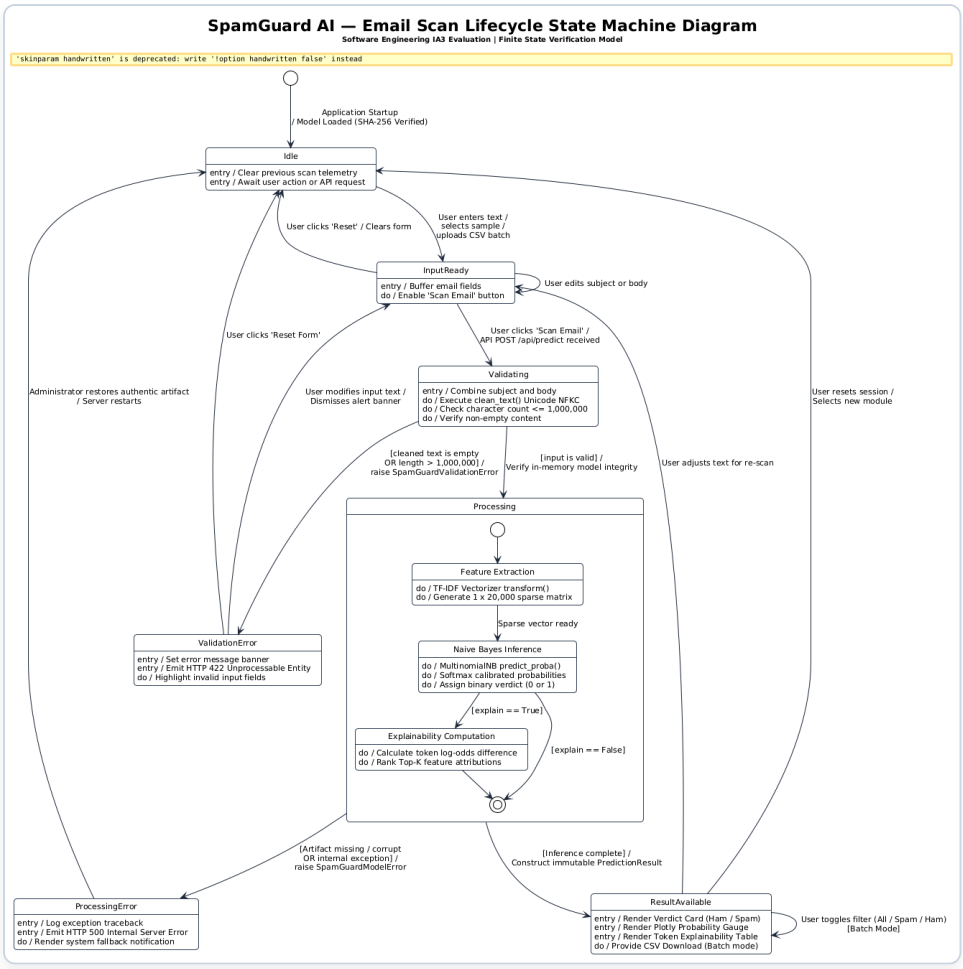
- **Source File:** `docs/ia3/uml/04_activity_diagram.puml`
- **Rendered Artifacts:** `04_activity_diagram.png` | `04_activity_diagram.svg`
- **Description:** Visualizes the sequential workflow and control logic of text classification: text combination, Unicode NFKC normalization, empty checks, payload length bounding (≤ 1,000,000 characters), cryptographic SHA-256 pre-inference gating, TF-IDF feature transformation, Naïve Bayes probability scoring, conditional log-odds explainability calculation, and UI result packaging.



SpamGuard AI Activity Diagram

Diagram 5: Email Scan Lifecycle State Machine Diagram

- **Source File:** `docs/ia3/uml/05_state_machine.puml`
- **Rendered Artifacts:** `05_state_machine.png` | `05_state_machine.svg`
- **Description:** Models the discrete operational states of an email scan entity: `Idle` \to `InputReady` \to `Validating` \to `Processing` (composite state with sub-states for Feature Extraction, Naïve Bayes Inference, and Explainability Computation) \to `ResultAvailable`. It also models defensive failure states (`ValidationError`, `ProcessingError`) and state recovery transitions.



SpamGuard AI State Machine Diagram